

How Do You Find My Touch? Mediating Social Support Through Vibrotactile Vest in the Context of Virtual Public Speaking

Yulia Sion
School of Digital Technologies
Tallinn University
Tallinn, Estonia
yzhigl@tlu.ee

Sunil Sudevan
Tallinn University
School of Digital Technologies
Tallinn, Estonia
sunilsudevan@gmail.com

David Lamas
School of Digital Technologies
Tallinn University
Tallinn, Estonia
david.lamas@tlu.ee



Figure 1: A speaker (left) delivering a virtual presentation while a supporter (right) provides remote tactile support during the public speaking activity.

Abstract

Touch is a powerful medium for emotional support, yet physical touch is often unavailable in remote settings. Mediated touch technologies offer a promising alternative, but their effectiveness depends on how tactile stimuli are interpreted and experienced. This paper explores abstract vibrotactile stimuli as a medium for social support in virtual public speaking. We conducted an exploratory study with eight dyads (eight speakers and eight supporters), in which speakers received non-representational vibrotactile patterns mapped from emotionally resonant musical sounds through a vibrotactile vest while delivering a virtual presentation. Our findings reveal that speakers associated the stimuli with encouragement, motivation, and attention redirection, though interpretations varied based on timing, intensity, and relational dynamics. Supporters found the system engaging, enabling them to express support in a

novel way. These insights contribute to designing mediated touch technologies by highlighting how abstract tactile patterns convey social meaning in remote contexts.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing devices.**

Keywords

vibrotactile, tactons, remote social support, public speaking

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1 Introduction

Touch is a fundamental mode of human communication, enabling the conveyance of affection, support, intimacy, and intent, often fostering social bonding. One of its most critical roles is in providing social support during stressful situations, where touch gestures

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such as a tap on the shoulder, a forearm caress, or a hug can effectively communicate reassurance, acceptance, and care [25, 26]. Prior research has shown that tactile support from close friends and family can mitigate stress effects [11] and serve as a protective mechanism against stress [25, 26].

However, physical touch is not always feasible due to situational constraints (e.g., presenting on stage) or physical distance (e.g., long-distance relationships). Technological advancements have sought to bridge this gap by simulating familiar tactile gestures, often leveraging vibrotactile feedback [9, 14, 19, 38]. Examples include wearable sleeves delivering pleasant stroking sensations to express care [16], devices simulating a tap on the shoulder to convey approval [4], and handheld devices enabling emotion transmission via squeezes [7]. However the difficulty in achieving realistic sensations, bulky forms, and cost make mediated real-life tactile sensations often impractical for daily use [37]. A subset of the mediated touch systems explores structured abstract tactile messages instead of simulating real-life sensations as a medium for conveying affective meaning in social interactions [6, 22]. Such approaches, instead of simulating realistic sensations, convey emotions and other social meanings through short patterns that represent concepts metaphorically or abstractly. The advantage of these forms of touches is their agility, versatility, and ability to convey complex patterns with 1-2 actuators [6]. Inspired by prior research demonstrating the emotional resonance of such abstract-mediated touches [22], our work investigates their potential as a medium for providing social support in stressful situations. Specifically, we focus on the context of public speaking, a common social evaluative scenario that induces significant stress for many individuals, often necessitating support from close friends or partners [12]. Despite the demonstrated need for mediated tactile support in such contexts, limited research has explored its usefulness for both the speaker and supporter in virtual public speaking scenarios. Addressing this gap, we pose the following research questions:

- RQ1: How do remote supporters and speakers experience mediated abstract tactile stimuli during virtual public speaking?
- RQ2: To what extent can abstract vibrotactile stimuli be interpreted as meaningful forms of social support?

To explore these questions, we conducted a study with eight dyads of close friends or partners, where one individual assumed the role of a speaker and the other as a supporter. Utilizing vibrotactile stimuli derived from emotionally resonant musical sounds, we examined the delivery of tactile feedback on the speaker's back. Our findings reveal that vibrotactile feedback effectively communicates varied meanings of social support, including encouragement, and motivation, and serves as an attention redirection mechanism. However, the type of stimuli, timing, and relational dynamics between senders and receivers significantly influence its effectiveness. These insights lay the groundwork for future confirmatory studies and provide practical implications for designing mediated tactile communication systems.

2 Related Work

This section provides an overview of prior work that informed and inspired this research. We focus on four key areas: the role of touch

in social support, mediated touch technologies for enabling social support, the design and perception of abstract tactile stimuli, and approaches to interpreting abstract tactile stimuli.

2.1 Touch and Social Support

Social support, as defined by Cutrona et al. [8], refers to the assistance provided through social bonds, encompassing instrumental, informational, and emotional forms of support. Emotional support, often expressed through touch, is particularly effective in reducing stress and anxiety. Morrison highlights the unique role of social touch as a “stress buffer” that fosters reassurance and care [26]. The most common meanings of emotional support include encouragement, praise, and emotional reflection [21]. Common supportive touch gestures, such as pats, squeezes, and spot touches, convey messages like “It’s okay” or “I’m here for you” [18]. Grewen et al. demonstrated that tactile contact prior to public speaking significantly reduced physiological stress indicators, emphasizing the critical role of touch in high-stress scenarios [29]. These findings underline the potential of tactile interventions to provide meaningful emotional support.

2.2 Mediated Touch for Social Support

The emergence of mediated touch technologies has focused on simulating real touch to foster presence, connectedness, and emotional expression in remote interactions [14, 38]. Early systems prioritized replicating realistic tactile sensations, with more recent advancements integrating mediated touch into VR environments for immersive experiences [3]. Notable examples include the Tactile Sleeve for Social Touch (TaSST) [15], EmoEcho [41], and SHIFTS [27], which utilize vibrotactile feedback to convey social cues and emotions. Research by Chan et al. [7] demonstrated that mediated touch in emotionally charged conversations could reduce stress indicators, highlighting its potential in stress-relief contexts. However, the pursuit of realism often necessitates bulky hardware and high costs, limiting the practicality of these systems for daily use.

A more feasible alternative involves abstract tactile messages or tactons [6], which encode and communicate social information through structured tactile patterns. For instance, Ju et al. [19] showed that short vibrotactile patterns could effectively represent emotional states like anger, joy, and relaxation. Similarly, Park et al.’s POKE device [28] allowed users to convey emotions during phone calls using bidirectional vibrotactile and force feedback. These approaches demonstrate the capability of tactons to convey emotions and intentions efficiently. However, their application in high-stress scenarios, such as public speaking, remains underexplored, presenting a significant opportunity for future research.

2.3 Designing Tactons for Social Support

Tactons leverage parameters such as frequency, amplitude, duration, waveform, and rhythmic patterns to convey information through the sense of touch [6]. Research shows that varying these parameters can elicit distinct affective responses [22, 33, 40, 43]. Particularly relevant to this work, studies on emotionally resonant sounds demonstrated that such sounds could generate tactons with corresponding emotional resonance [10, 13, 22, 34]. Building on this

foundation, our prototype employs emotionally resonant vibrotactile patterns inspired by musical sounds to convey social support.

2.4 Interpreting Tactons in The Context of Social Support

Verbeek and Ihde's [36] concepts of hermeneutic and embodied mediation offer frameworks for understanding how mediated tactile communication is interpreted. Hermeneutic mediation involves an interpretive step, where users consciously decode abstract representations, whereas embodied mediation fosters direct, pre-reflective experiences [32, 44].

The prior research highlights the potential of tactons in fostering connection and providing social support [4, 41]. However, there is limited research on their application in high-stress scenarios.

In designing vibrotactile patterns for social support, we sought to prioritize embodied experiences, enabling users to intuitively perceive and associate sensations with supportive meanings. For instance, a pleasant vibrational pattern should immediately evoke encouragement or reassurance without requiring conscious interpretation. We hypothesize that emotionally resonant stimuli enhance embodied mediation, facilitating more intuitive and effective communication. By synthesizing insights from these areas, this study extends prior work by exploring how vibrotactile feedback can be designed to provide meaningful social support in high-stress scenarios, such as public speaking.

3 Method

This exploratory study was conducted in a semi-controlled environment simulating a virtual public speaking scenario. Controlled elements included the speech duration and the timing of mediated touch delivery (before, during, and after the speech).

3.1 Apparatus

The vibrotactile output system consisted of a custom vest made of stretchable material, incorporating six voice-coil actuators (2x3 grid; TEAX13C02, Tectonic Elements). These actuators, measuring 31.75mm in diameter and 9mm in height, were embedded in the upper rear of the vest to target the participant's back. The actuators offered a broad frequency range (50 Hz–2 kHz) and were chosen for their compact size and resonance frequency of 560 Hz. The back was selected as the feedback site based on prior research demonstrating its suitability for conveying social support [18].

The system was controlled via an ESP8266 board interfaced with an MP3 Module (DF Mini Player), amplifiers (MAX98306), and actuators. The MP3 files were normalized to 89dB using Mp3Gain. Stimulus control and data logging were managed over Wi-Fi using Arduino Cloud and CoolTerm, respectively.

3.2 Stimuli

Stimuli consisted of vibrotactile patterns mapped from 10-second music clips sourced from the DEAM library [1], which provides compositions validated for their affective properties, with assigned valence and arousal scores. Eight distinct compositions, representing opposite quadrants of the Circumplex Model of Affect [39], were mapped to eight vibrotactile patterns (VB1-VB8). The primary criteria for selecting the stimuli was to have a variety of sensations

(from highly pleasant to unpleasant, and from highly intense to very relaxing) so that supporters could have enough variations to assign different meanings to the proposed stimuli.

The mapping process, validated in our prior study [34], ensured that the tactile patterns elicited emotional responses aligned with their auditory counterparts. This audio-tactile mapping approach, inspired by previous research, builds on evidence that emotionally resonant sounds can evoke similar affective responses when translated into vibrotactile stimuli [10, 13, 22]. Table 1 in the Appendix presents the selected stimuli, along with their corresponding valence and arousal scores.

3.3 Participants

Eight dyads (16 participants) were recruited, including two pairs of friends (1–12 years relationship duration) and six family pairs (five partners, one pair of siblings). Participants (5 males, 11 females) ranged in age from 20 to 39 years ($M=25$). Roles were self-assigned: three males and five females acted as Speakers (SP), and the remaining participants as Supporters (SU). Ethical approval was obtained, and all participants provided written consent.

3.4 Procedure

The study consisted of four phases: preparation, stimulus selection, public speaking, and post-study interviews (Figure 2).

- (1) Preparatory Stage: Speakers prepared 6-minute presentations in English (Pecha Kucha format) on their chosen topic. Upon arrival, dyads were separated. Speakers completed the PRCA-24 Communication Apprehension Measure [23], while Supporters completed the Toronto Empathy Questionnaire [35]. Both completed demographic surveys covering relationship closeness, age, and English proficiency.
- (2) Stimulus Selection: Speakers wore the vest and adjusted the fit. After testing the vibrations, they evaluated eight stimuli, assigning meanings based on five emotional support categories from prior research (provided to them on a printed paper) [21]. Speakers selected three preferred stimuli for the public speaking task. These were then shared with Supporters, who could test but not modify the stimuli or their assigned meanings.
- (3) Public Speaking Task: The activity was conducted via Zoom. Speakers, wearing the vest and noise-canceling headphones, presented to a virtual audience of 2–3 unknown individuals. Supporters, located in a separate room, observed the session and provided mediated touch support using a laptop interface with three buttons representing the selected stimuli. Supporters could send stimuli freely during a 2-minute pre-speech period, the 6-minute speech, and a 2-minute post-speech phase. Two researchers ensured smooth execution of the study.
- (4) Post-Study Interviews: Dyads participated in a semi-structured interview to discuss their experiences. Topics included the usefulness and emotional impact of tactile stimuli. Surveys assessed stress levels, perceived disturbance, and overall feedback.

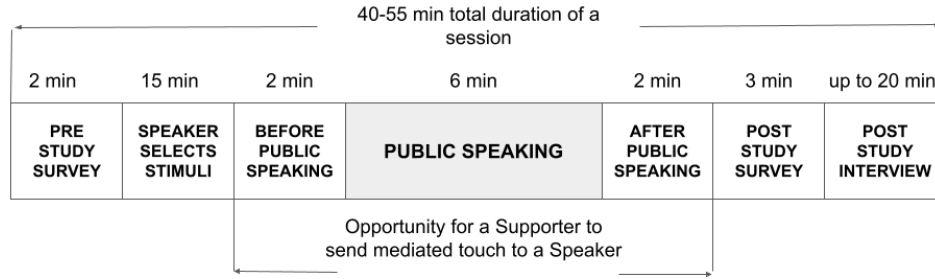


Figure 2: The schematic representation of the study procedure.

3.5 Data Collection and Analysis

Quantitative data included logged stimulus delivery times, pre- and post-study surveys, and video recordings. Facial expressions and body posture during the Zoom sessions were analyzed using the Evaluated Speech Performance Measure (ESPM) [31], focusing on speech volume, continuity, body tension, fidgeting, and smiling. Video analysis (counting frequency of each instance from the ESPM) involved two independent reviewers validating observations. The post-study survey was adapted from the validated questionnaire for Measuring the Social Disfidence of Mediated Social Touch Technologies [24]. Qualitative data were derived from transcribed interviews, coded by two researchers using Thematic Analysis [5] with Affinity Diagramming [30]. Initial codes aligned with research questions, while subsequent iterations identified emergent themes.

4 Findings

The Thematic Analysis and quantitative data analysis revealed four main themes: Usefulness of Mediated Tactile Support, Experience of Sending Mediated Tactile Support, and Interpretation and Perception of Vibrotactile Stimuli.

4.1 Usefulness of Mediated Tactile Support

Participants revealed differing preferences and perceptions regarding the timing and usefulness of mediated tactile support. Pre-speech touch was generally appreciated, with six (out of eight) speakers describing it as pleasant and motivating. For instance, SP1 shared, “Before the presentation, she sent me the vibration ‘you can do it.’ It made me feel good and motivated.” SP5 noted, “It felt like someone was there.” However, two participants, SP6 and SP8, stopped pre-speech vibrations, citing: “The vibrations before starting made me feel anxious.”

During the speech, four participants found the stimuli pleasant, one found it unpleasant, and three were neutral. Half of the participants agreed it improved their focus or performance. SP4 described it as “disturbing” but admitted, “The one I stopped during the speech helped me slow down because I felt I was going fast.” SP1 mentioned how touch helped her refocus when nervous, stating, “It made me stop thinking I wasn’t doing a good job.” SP5 highlighted the usefulness of mediated touch as simulated audience feedback: “It felt like someone was paying attention.”

Post-speech vibrations were universally praised, with most participants comparing them to “applause” or a “relief.” SP8 described this phase as “100 percent the best time to receive the vibration—it

felt like success.” Similarly, SP3 stated that post-speech touch “helped to relax.”

Observational data supported these findings. Participants who appreciated the touch often smiled or laughed during the pre-speech phase. During the speech, some (participants who appreciated receiving the stimuli) paused or adjusted their posture when receiving vibrations, while others (participants who found stimuli disturbing) showed signs of nervousness, such as fidgeting. These behaviors aligned with their feedback on whether the touch was perceived as supportive or distracting.

4.2 Experience of Sending Mediated Tactile Support

Supporters generally felt that mediated touch improved the speakers’ performance. Seven (out of eight) supporters agreed that they could accurately convey their intentions through tactile stimuli. The timing logs showed peaks in vibrations during the second, fourth, seventh, and final minutes of the speech, with the highest activity occurring just after the speech concluded. Supporters tailored their use of vibrations to the speaker’s needs. SU3 observed, “When I sent the vibration, his facial expression changed, and I felt more connected to him.” On the other hand, some supporters expressed concerns about making speakers uncomfortable. SU6 remarked, “I worried about whether the vibration would be too strong, but seeing her reaction helped me adjust.” Others appreciated how wearing the vest themselves allowed them to empathize with the speaker’s experience. For example, SU2 said, “Feeling the same vibration made it easier to understand how my partner felt.” Post-speech touch was appreciated by all supporters, with SU5 describing it as “a rewarding way to communicate support.”

4.3 Interpretation and Perception of Vibrotactile Stimuli

Speakers demonstrated varying abilities to interpret and recall the meanings of vibrations. Half could recall the specific meanings associated with vibrations after the speech, while others struggled, citing stress or focus on the task. SP1 noted, “I don’t remember specific meanings, but I could sense whether the vibrations were positive or negative.” Mapping of vibrations to emotional support categories revealed consistent alignment with their original affective properties. For instance, VB2, associated with Directed Action, was generated from a sound with moderate arousal and low valence. Similarly, VB1, linked to Praise and Reassurance, was derived

from high valence, high arousal sounds. Participants' qualitative descriptions reinforced these mappings. SP6 found VB6 to be "calm" and "pleasant," while SP4 described VB1 as "strong, like someone clapping." The least preferred vibrations (VB4 and VB5) were generated from low valence, high arousal sounds, aligning with their intended unpleasant properties.

5 Discussion

This study explored the experiences of senders and receivers of abstract-mediated tactile support during virtual public speaking and examined the extent to which abstract vibrotactile stimuli can be interpreted as meaningful forms of social support. Below, we address the two key research questions in light of our findings and related work.

- Usefulness of Remote Tactile Support for Public Speaking

Tactile Stimuli as Attention Redirection and Performance Enhancement. Participants described tactile stimuli as effective tools for redirecting attention during moments of distraction or nervousness. For example, SP1 noted, "The vibration helped me stop thinking I wasn't doing a good job and refocus on my speech." This aligns with cognitive-behavioral techniques, where competing stimuli redirect focus from stress to a reflective state [17, 20], and studies on attention management [2, 42]. While not explicitly designed for attention management, the tactile cues in our study provided this additional benefit, enhancing focus and motivation. Some participants also interpreted vibrations as corrective feedback, prompting adjustments in speech pacing. SP4 shared, "The vibration helped me realize I was going too fast, so I slowed down." These findings suggest abstract tactile support not only fosters emotional reassurance but also improves performance by increasing situational awareness.

Motivation and Social Presence. The utility of vibrations was influenced by timing. Pre-speech stimuli were largely reassuring, while mixed responses during the speech suggest the need for subtle, minimally intrusive patterns. Post-speech feedback was universally celebrated, suggesting potential for energetic, affirmational designs. The relational dynamics between sender and receiver amplified these effects, with participants often attributing the emotional impact of the vibrations to their knowledge of the sender's identity. This finding echoes research by Chan et al. [7], which demonstrated the emotional value of tactile cues in stressful contexts and the overall benefits of touch in providing emotional support and a feeling of connectedness [25, 26]. Our findings underline the importance of tailoring tactile interventions to individual preferences and situational contexts. Designers should consider adaptive systems capable of modulating intensity, timing, and pattern complexity based on user feedback or contextual cues. For example, subtle vibrations might be more effective during high-focus activities, while more pronounced patterns could enhance post-event recognition and affirmation. Future designs should also leverage the relational dynamics by allowing users to co-create or personalize tactile messages, thereby enhancing the emotional connection and sense of shared presence.

- Interpretation of Abstract Vibrotactile Stimuli

General and Specific Interpretations. Participants varied in their ability to recall specific meanings of vibrations, with many describing them in broader terms, such as "positive" or "negative." This suggests that abstract tactile stimuli effectively communicate emotional valence without requiring detailed interpretation, aligning with Verbeek and Ihde's theory of embodied mediation, where tactile feedback fosters intuitive, low-effort understanding [36]. The findings suggest a balance between hermeneutic and embodied interpretations. While specific recall required conscious effort, the embodied experience was sufficient for conveying emotional support, highlighting the flexibility of abstract vibrotactile stimuli for diverse user abilities to recall meaning.

Emotional Resonance through Sound-Based Design. The consistent alignment between vibrotactile stimuli and their affective properties demonstrates the potential of emotionally resonant stimuli as a design material. For instance, vibrations derived from high-valence, high-arousal sounds were universally associated with praise and motivation, and low-valence stimuli were perceived as a corrective nudge. These findings validate the use of sound-to-touch mappings as a viable approach for designing tactile communication systems that encode nuanced emotional meanings [22, 34].

6 Future Direction

This study highlights several opportunities for advancing mediated tactile support systems. While our findings provide valuable insights, the limited sample size (eight dyads) constrains the generalizability of the results. In future studies we plan to recruit larger and more diverse participant groups, employing between-subject designs to further examine the impact of mediated tactile support on stress indicators across various contexts. To substantiate the effects of tactile support on stress reduction, future work could integrate physiological measures, such as heart rate variability and skin conductance, alongside subjective assessments. Field studies in naturalistic environments would also validate the practicality and impact of these systems outside controlled lab settings. Several participants expressed interest in broader vibrotactile stimuli and alternative form factors, such as wristbands, to improve usability.

7 Conclusion

This exploratory study contributes to the understanding of how mediated touch can serve as a meaningful channel for social support during virtual public speaking. By investigating dyadic interactions between senders and receivers, we reveal how abstract vibrotactile stimuli convey encouragement, motivation, and attention, enhancing both performance and emotional connection. Our findings underscore the influence of timing, relational dynamics, and tactile patterns on the perceived utility and interpretation of mediated touch. Through an exploratory design leveraging emotionally resonant musical patterns, we demonstrate the potential of vibrotactile feedback as a versatile and effective medium for communication in stressful contexts. This work lays the groundwork for designing more intuitive, flexible, and contextually adaptive tactile systems that provide emotional and functional support in high-stress scenarios.

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A Stimuli Data

A.1 Provided Vibrotactile Stimuli

Table 1 below presents vibrotactile stimuli offered to speakers for selection and meaning assignment. Each stimulus has an associated valence and arousal score, which were assigned based on the sounds from which the tactile stimuli were derived. These scores were sourced from the DEAM library [1].

Stimulus ID	Valence score	Arousal score
VB1	6.1	6.1
VB2	3.6	3.3
VB3	5.6	4.3
VB4	4.4	5.4
VB5	2.2	2.7
VB6	3	5
VB7	5.8	2.9
VB8	7	8.1

Table 1: Tactile stimuli and corresponding arousal and valence scores.

A.2 Social Support Categories

Table 2 presents social support categories and their meanings, as derived from the study by Kindness et al [21]. Supporters could assign these predefined meanings to the tactile stimuli or suggest their own.

Social Support Category	Meaning
Directed Action	Do not rush
Directed Action	Take a deep breath
Directed Action	Stay focused
Directed Action	Try a little harder
Emotional Advice	Stay positive
Emotional Reflection	I know how you feel
Praise	Good job
Reassurance	It is going to be fine
Reassurance	You can do this

Table 2: Social Support Categories and Their Meanings for Assigning Vibrations.

B Qualitative Analysis of Stimulus-to-Meaning Mapping

Figure 3 illustrates how selected stimuli were mapped to their assigned meanings.

C Qualitative Analysis about the Speakers' Perception of the Stimuli

The bar charts below (Figures 4 - 10) summarize responses from the post-study questionnaire, which evaluated supporters' subjective perceptions of the tactile stimuli in terms of disturbance and relaxation before, during, and after the public speaking activity. The descriptive statistics also illustrate the popularity of social support categories and their associated meanings as selected by the speakers.

D Qualitative Analysis of Supporters' Experience

Supporters were asked whether they perceived the tactile stimuli as potentially disturbing for the speakers (Figure 11). Figure 12 depicts the frequency of stimuli sent by supporters before, during, and after the speech.

Emotional Support categories mapped to the vibration stimuli.

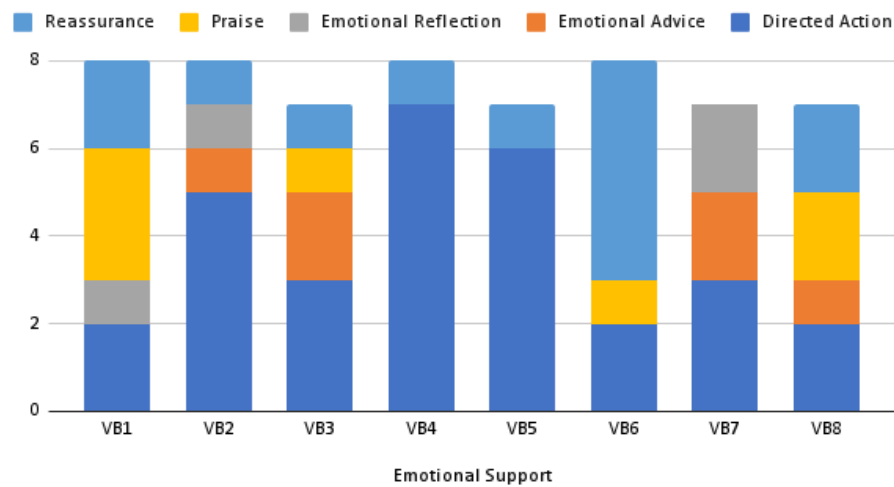


Figure 3: Speakers' Assignment of Vibrotactile Stimuli to Emotional Support Categories.

Receiving the mediated touch before my speech was disturbing.

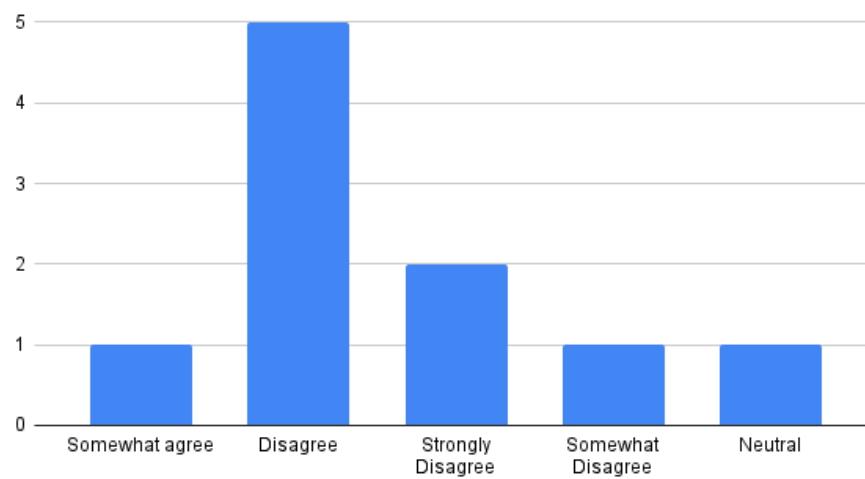


Figure 4: Self-report on whether the mediated touch was perceived as disturbing when sent before the speech.

Receiving the mediated touch during my speech was disturbing.

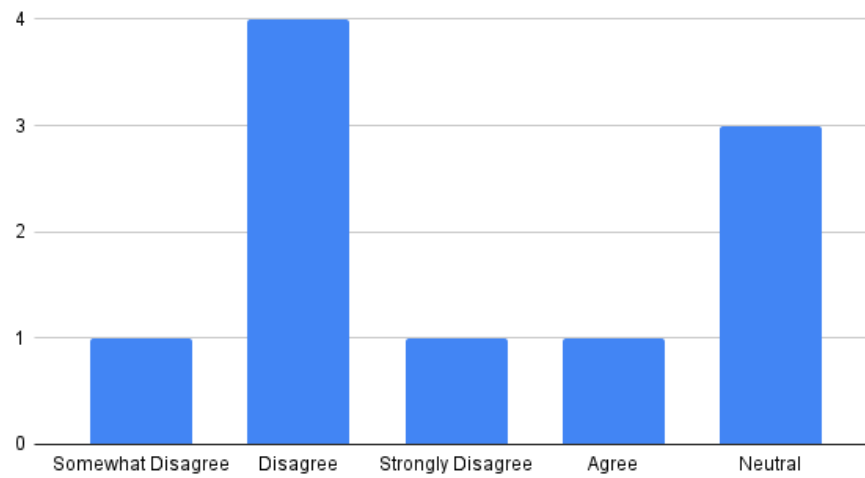


Figure 5: Self-report on whether the mediated touch was perceived as disturbing when sent during the speech.

Receiving the mediated touch after my speech was disturbing.

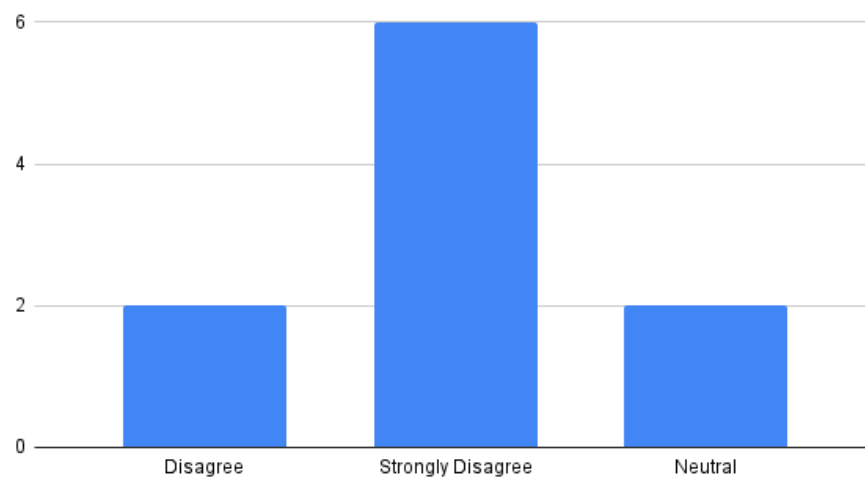


Figure 6: Self-report on whether the mediated touch was perceived as disturbing when sent after the speech.

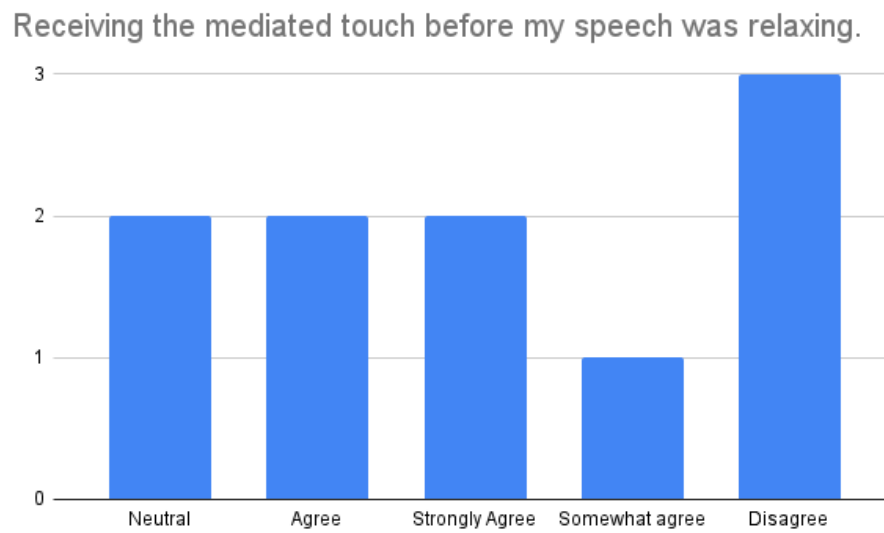


Figure 7: Self-report on whether the mediated touch was perceived as relaxing when sent before the speech.

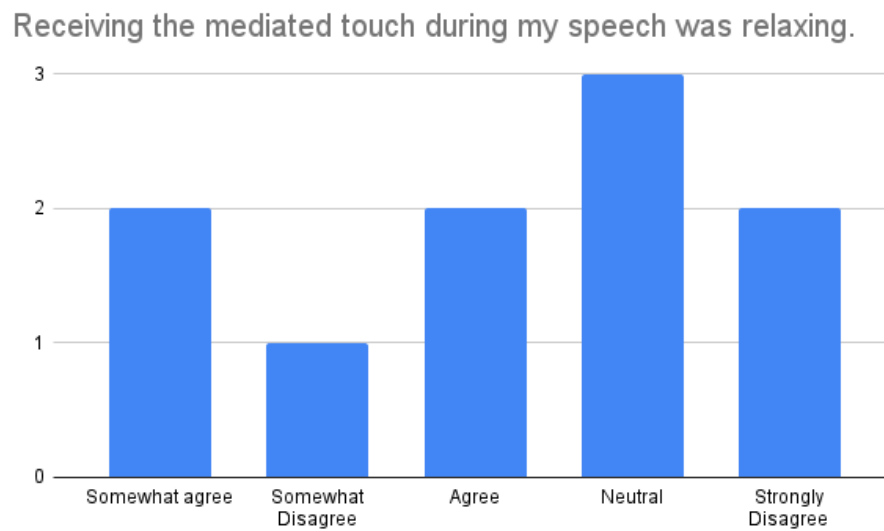


Figure 8: Self-report on whether the mediated touch was perceived as relaxing when sent during the speech.

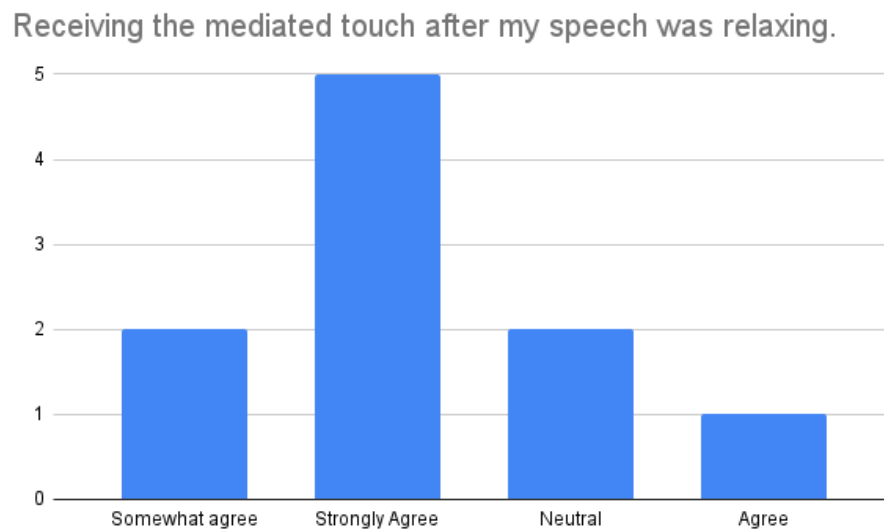


Figure 9: Self-report on whether the mediated touch was perceived as relaxing when sent after the speech.

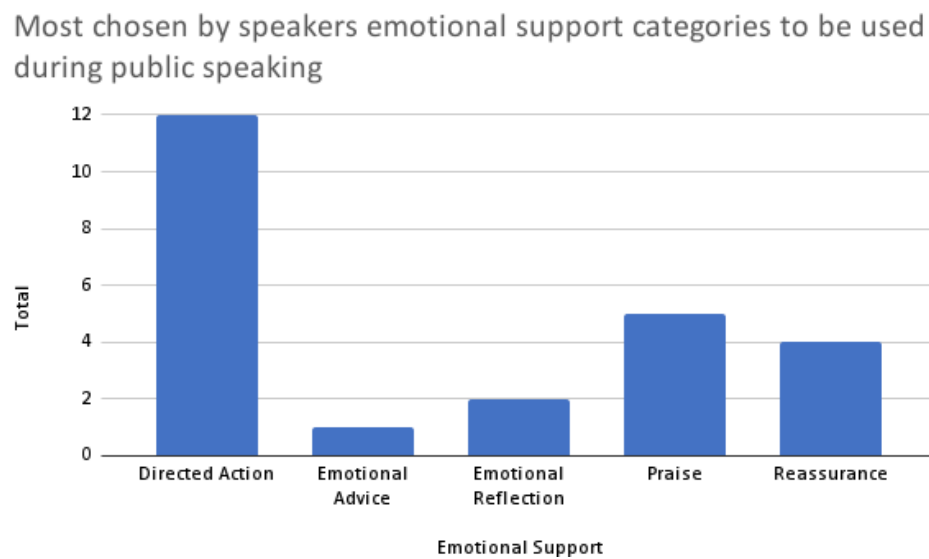


Figure 10: Emotional Support Categories Selected by Speakers for Use by Their Supporters During Public Speaking.

Receiving the mediated touch was useful in improving the quality of my partner's presentation.

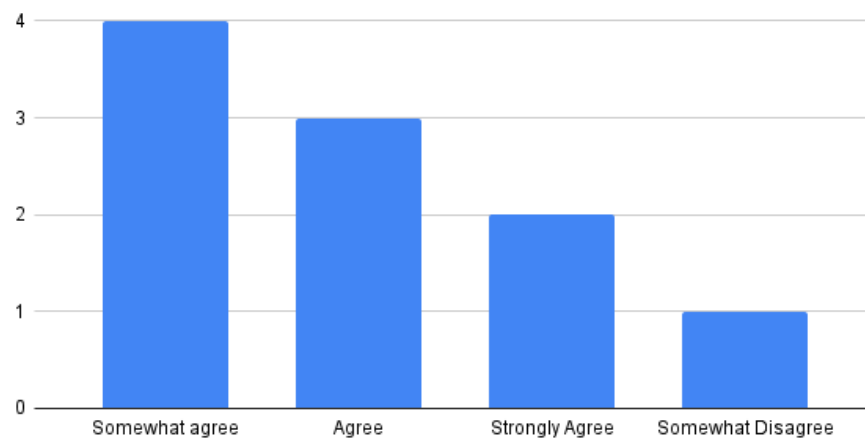


Figure 11: Supporters' Perceptions of Mediated Touch in Enhancing Their Speakers' Public Speaking.

Total sent support on a timeline

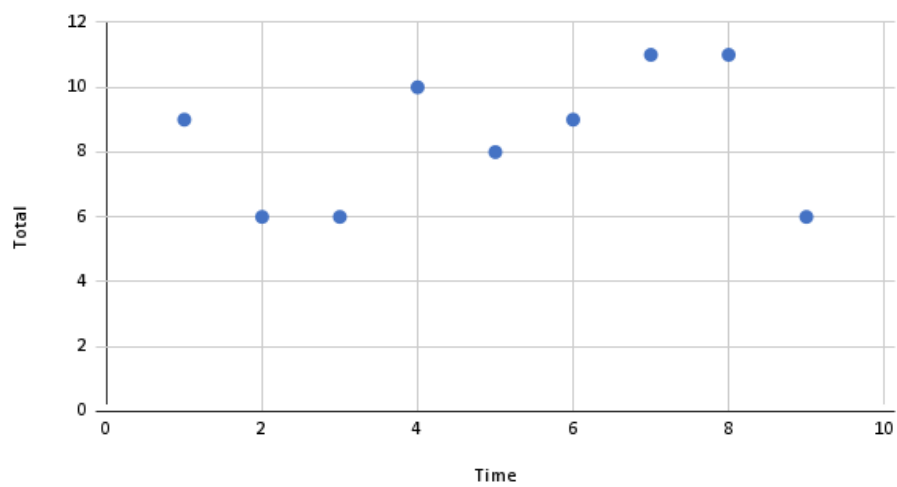


Figure 12: Total Number of Stimuli Provided by Supporters During Public Speaking, Including Pre and Post Speech Periods.